

DNA & Protein Sequence Retrieval and BLAST Analysis; Automatic and By Hand

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1. Introduction

Public health repositories such as NCBI and UniProt are now filled with hundreds of millions of data sequences making the ability to retrieve a certain sequence of interest and comparing it against other sequences fundamental to every branch of biological data analysis. The Basic Local Alignment Search Tool (BLAST). However, performing this analysis manually becomes inefficient when tasks must be repeated across many sequences and integrated within a larger framework or reproduced by other researchers. These concerns motivated the development of an end to end framework in Python programming language that automated the full pipeline from downloading a FASTA file to BLAST submission and parsing the information and extracting it from a long text file.

2. Background

2.1 Sequence Databases and Access

NCBI handles many sequencing related databases like RefSeq, GeneBank and Protein. They are easily accessible through the GUI interface of the official NCBI database (<https://www.ncbi.nlm.nih.gov/search/?term=>). NCBI database also supports fetching information and FASTA file from your local terminal via Entrez Programming Utilities (E Utilities). UniProt, by contrast offers curated protein sequence with rich functional annotation on through a REST API.

2.2 BLAST and Alignment Metrics

BLAST is a search algorithm that identifies local alignments by firstly locating the short high scoring word hits. According to the NCBI BLAST documentation percent identity refers to the fraction of position with identical residues, percent of similarity counts conservative substitutions and the bit score is the raw alignment score normalized for comparison across searches. It is important to notice none of these metrics alone can contribute to the realisation of homology which is an evolutionary inference.

3. Methodology

To perform the given task. I first accessed the databases from their GUI and downloaded a sample FASTA file, then copied the full sequence of Gene/Protein inside BLAST (BLASTn for nucleotide and BLASTp for amino acids) and performed the analysis which is easily traceable through the screenshots. Later, for better robustness a jupyter notebook was written to implement a three staged pipeline. 1. Secure access to data and download FASTA file through HTTP GET or Entrez fetch. Then, 2. Each sequence is submitted to BLAT with qblast. 3. The XML result is parsed and the data should be extracted leveraging string functions and finally, outputted as a CSV file.

4. Limitations

Through the implementation of this pipeline, the function handling the analysis failed multiple times. Therefore, I assigned an extra EBI engine, error handling section along with execution limit time for the function. Also, it is important to notice that the accessibility to NCBI databases is through a KEY which will be bound to your accounts email to track and limit the requests from certain accounts.